

Then it holds that

- The function f has a unique fixed point $p \in D$.
- For any initial point $x^{(0)} \in D$, the fixed point iteration $x^{(t)} = f(x^{(t-1)})$, $t \geq 1$, converges to the fixed point p as $t \rightarrow \infty$.
- The following error bounds hold:

$$\begin{aligned}\|x^{(t)} - p\|_\infty &\leq \frac{K^t}{1-K} \|x^{(1)} - x^{(0)}\|_\infty, \\ \|x^{(t)} - p\|_\infty &\leq \frac{K}{1-K} \|x^{(t)} - x^{(t-1)}\|_\infty, \\ \|x^{(t)} - p\|_\infty &\leq K \|x^{(t-1)} - p\|_\infty.\end{aligned}$$

Proof. We only need to show that f is a contraction. For any $y, y' \in \mathbb{R}^d$, for any $i \in [d]$, there exists z on the line segment between y and y' such that

$$\begin{aligned}|f_i(y) - f_i(y')| &\leq \left| \left\langle \frac{df_i(x)}{dx} \Big|_{x=z}, y - y' \right\rangle \right| \\ &\leq \left\| \frac{df_i(x)}{dx} \Big|_{x=z} \right\|_1 \|y - y'\|_\infty \\ &\leq K \|y - y'\|_\infty.\end{aligned}$$

where the first step follows from mean value theorem, the second step uses the Holder's inequality, and the last step is due to the second condition.

Therefore, by definition of $\|\cdot\|_\infty$, we have

$$\|f(y) - f(y')\|_\infty \leq K \|y - y'\|_\infty.$$

Hence f is contractive with contractivity constant $K \in [0, 1)$. By Lemma A.3, the proof is complete. $\square$

A.4 Matrix Case

In this section, we extend the Banach fixed point theorem to its matrix version. Note that $\mathbb{R}^{n \times d}$ is a Banach space with norm $\|\cdot\|_\infty$. The maximum norm of $A \in \mathbb{R}^{n \times d}$ is defined as $\|A\|_\infty = \max_{i \in [n], j \in [d]} |A_{i,j}|$.

Lemma A.6 (Banach fixed point theorem). *Let $D \subseteq \mathbb{R}^{n \times d}$ be a nonempty closed set. Suppose that $f : D \rightarrow \mathbb{R}^{n \times d}$ is differentiable and satisfies the following:*

- $f(X) \in D$ whenever $X \in D$.
- There exists constant $K < 1$ such that for every $i, k \in [n]$ and $j, l \in [d]$

$$\left| \frac{df(X)_{i,j}}{dX_{k,l}} \right| \leq \frac{K}{nd}, \quad \forall X \in D.$$

Then it holds that

- The function f has a unique fixed point $p \in D$.

- For any initial point $x^{(0)} \in D$, the fixed point iteration $x^{(t)} = f(x^{(t-1)})$, $t \geq 1$, converges to the fixed point p as $t \rightarrow \infty$.
- The following error bounds hold:

$$\begin{aligned} |x^{(t)} - p| &\leq \frac{K^t}{1-K} |x^{(1)} - x^{(0)}|, \\ |x^{(t)} - p| &\leq \frac{K}{1-K} |x^{(t)} - x^{(t-1)}|, \\ |x^{(t)} - p| &\leq K |x^{(t-1)} - p|. \end{aligned}$$

Proof. Here we only need to show that f is a contraction. For any $X, X' \in \mathbb{R}^{n \times d}$, for any $i \in [n], j \in [d]$, there is a Z between the matrix X and X' such that

$$\begin{aligned} |f(X)_{i,j} - f(X')_{i,j}| &\leq \left| \frac{df(X)_{i,j}}{dX} \right|_{X=Z} \circ (X - X')| \\ &\leq \left| \frac{K}{nd} (\mathbf{1}_{n \times d} \circ (X - X')) \right| \\ &\leq K \|X - X'\|_{\infty} \end{aligned}$$

where the first step follows from mean value theorem, the second step uses the second condition in the lemma, and the last step is due to basic algebra.

Therefore, by definition of $\|\cdot\|_{\infty}$, we have

$$\|f(X) - f(X')\|_{\infty} \leq K \|X - X'\|_{\infty}$$

Hence f is contractive with contractivity constant $K \in [0, 1)$. By Lemma A.3, the proof is complete. $\square$

A.5 Examples Using Fixed Point Theorem

In this section, we present some calculation case by using Banach fixed point theorem.

Theorem A.7. *Fixed Point Theorem:*

- If $g \in C[a, b]$ and $a \leq g(x) \leq b$ for all $x \in [a, b]$, then g has least one fixed point in $[a, b]$
- If, in addition, g' exists in $[a, b]$, and $\exists k < 1$ such that $|g'(x)| \leq k < 1$ for all x , then g has a unique fixed point in $[a, b]$.

Success Case: $g(x) = \frac{x^2-1}{3}$ has a unique fixed point in $[-1, 1]$.

Proof. First we need show $g(x) \in [-1, 1]$, $\forall x \in [-1, 1]$. Find the max and min values of g as $-\frac{1}{3}$ and 0. (Hint: find critical points of g first). So $g(x) \in [-\frac{1}{3}, 0] \subset [-1, 1]$.

Also $|g'(x) = \frac{2x}{3}| \leq \frac{2}{3} < 1$, $\forall x \in [-1, 1]$, so g has unique fixed point in $[-1, 1]$ by Fixed Point Theorem. $\square$

Failed Case 1: $g(x) = \frac{x^2-1}{3}$ has a unique fixed point in $[3, 4]$. But we can't use FPT to show this.

Note that there is a unique fixed point in $[3, 4]$ ($p = \frac{3+\sqrt{13}}{2}$), but $g(4) = 5 \notin [3, 4]$, and $g'(4) = \frac{8}{3} > 1$ so we cannot apply FPT here.

From this example, we know FPT provides a sufficient but not necessary condition.

Failed Case 2: We can use FPT to show that $g(x) = 3^{-x}$ must have Fixed Point on $[0, 1]$, but we can't use FPT to show if it's unique (even though the FP on $[0, 1]$ is unique in this example).

Solution. $g'(x) = (3^{-x})' = -3^{-x} \ln 3 < 0$, therefore $g(x)$ is strictly decreasing on $[0, 1]$. Also $g(0) = 3^0 = 1$ and $g(1) = 3^{-1}$, so $g(x) \in [0, 1]$, $\forall x \in [0, 1]$. So a FP exists by FPT.

However, $g'(0) = -\ln 3 \approx -1.098$, so we don't have $|g'(x)| < 1$ over $[0, 1]$. Hence FPT does not apply.

Nevertheless, the FP must be unique since g strictly decreases and intercepts with $y = x$ line only once.

Then We will present a lemma that proves a quadratic polynomial function cannot have two fixed points that satisfy the conditions of the Banach fixed point theorem.

Lemma A.8. *There doesn't exist a quadratic function that contains two fixed points and simultaneously satisfies that points in the neighborhood of the fixed points can converge to the respective fixed point through iteration.*

Proof. For $\forall a, b, c \in \mathbb{R}$, let $f(x) = ax^2 + bx + c$ be a quadratic function. Then we have $f'(x) = 2ax + b$. Assuming that $f(x)$ has two fixed points, then we have:

$$x_1 = \frac{(1-b) + \sqrt{(b-1)^2 - 4ac}}{2a}, x_2 = \frac{(1-b) - \sqrt{(b-1)^2 - 4ac}}{2a}$$

Then we have,

$$f'(x_1) = 1 + \sqrt{(b-1)^2 - 4ac} > 1$$

$$f'(x_2) = 1 - \sqrt{(b-1)^2 - 4ac} < 1$$

So there is only one fixed point can satisfy the second condition of Lemma A.3. □

B CASE STUDY: POLYNOMIAL ACTIVATION

In this section, we start with a simple example of a single variable function which has at least two fixed points that can be found using the fixed point method. Then we use it to construct a specific neural network with polynomial activation function. Finally, we show that this neural network has at least two fixed points that can be found using the fixed point method.

B.1 Polynomial Activation

Firstly, we present a univariate polynomial function that has at least two fixed points, which can be identified using the fixed point method.

Lemma B.1. *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined as $f(x) := -\frac{2}{5}x^4 + \frac{3}{2}x^2$. Then the following statements hold:*

- *The function f has at least two fixed points which can be found by the fixed-point iteration, and we denoted them as p_1, p_2 .*
- *For $i \in \{1, 2\}$, there exists a constant $\epsilon_i > 0$, for any initial point $x^{(0)} \in [p_i - \epsilon_i, p_i + \epsilon_i]$, the fixed-point iteration $x^{(t)} = f(x^{(t-1)})$ converges to the fixed point p_i .*

- For $i \in \{1, 2\}$, there exists a constant $c_i > 0$ and a constant $K_i \in [0, 1)$ such that for any $t \geq 2$,

$$|x^{(t)} - p_i| \leq K_i^t \cdot c_i \epsilon_i.$$

Proof. Clearly, $p_1 = 0$ and $p_2 \approx 1.4028$ are two fixed points of f . We show that p_1 and p_2 can be found by the fixed-point iteration and the error bounds hold.

For the fixed point $p_1 = 0$, let $\epsilon_1 = 0.3$, $K_1 = 0.9$, and $C_1 = 2/(1 - K_1)$. For any $x \in [p_1 - \epsilon_1, p_1 + \epsilon_1] = [-0.3, 0.3]$, we have $f(x) \in [0, 0.132] \subseteq [-0.3, 0.3]$. Hence the first condition of Lemma A.4 is satisfied. For any $x \in [p_1 - \epsilon_1, p_1 + \epsilon_1]$ we have $|f'(x)| < K_1 = 0.9 < 1$. Hence the second condition of Lemma A.4 is satisfied. Thus by Lemma A.4, if we pick the initial point $x^{(0)} \in [p_1 - \epsilon_1, p_1 + \epsilon_1]$, we can find p_1 by the fixed point iteration $x^{(t)} = f(x^{(t-1)})$, and it holds that for any $t \geq 2$,

$$\begin{aligned} |x^{(t)} - p_1| &\leq \frac{K_1^t}{1 - K_1} |x^{(1)} - x^{(0)}| \\ &\leq \frac{K_1^t}{1 - K_1} \cdot 2\epsilon_1 \\ &= K_1^t \cdot C_1 \epsilon_1, \end{aligned}$$

where the first step follows from Lemma A.4, the second step follows from $x^{(0)}, x^{(1)} \in [p_1 - \epsilon_1, p_1 + \epsilon_1]$, and the last step follows from $c_1 = 2/(1 - K_1)$.

For the fixed point $p_2 \approx 1.4028$, let $\epsilon_2 = 0.1$, $K_2 = 0.92$, and $C_2 = 2/(1 - K_2)$. For any $x \in [p_2 - \epsilon_2, p_2 + \epsilon_2] = [1.3028, 1.5028]$, we have $f(x) \in [1.347, 1.397] \subseteq [1.3028, 1.5028]$. Hence the first condition of Lemma A.4 is satisfied. For any $x \in [p_2 - \epsilon_2, p_2 + \epsilon_2]$, we have $|f'(x)| < K_2 = 0.92 < 1$. Hence the second condition of Lemma A.4 is satisfied. Thus by Lemma A.4, if we pick the initial point $x^{(0)} \in [p_2 - \epsilon_2, p_2 + \epsilon_2]$, we can find p_2 by the fixed point iteration $x^{(t)} = f(x^{(t-1)})$, and it holds that for any $t \geq 2$,

$$\begin{aligned} |x^{(t)} - p_2| &\leq \frac{K_2^t}{1 - K_2} |x^{(1)} - x^{(0)}| \\ &\leq \frac{K_2^t}{1 - K_2} \cdot 2\epsilon_2 \\ &= K_2^t \cdot c_2 \epsilon_2, \end{aligned}$$

where the first step follows from Lemma A.4, the second step follows from $x^{(0)}, x^{(1)} \in [p_2 - \epsilon_2, p_2 + \epsilon_2]$, and the last step follows from $c_2 = 2/(1 - K_2)$.

Therefore, we complete the proof. $\square$

Next, we prove two lemmas which are useful to construct a neural network with a polynomial activation function.

Lemma B.2. *Let*

$$f_1(x) := -\frac{2}{5}\langle u, x \rangle^4 + \frac{8}{5}C\langle u, x \rangle^3 + \left(\frac{3}{2} - \frac{12}{5}C^2\right)\langle u, x \rangle^2 + \left(\frac{8}{5}C^3 - 3C\right)\langle u, x \rangle + 1,$$

where $C := \sqrt{\frac{15+\sqrt{65}}{8}}$, $u := [1, 1, C - 1]$. Let $f_2(x) := 1, f_3(x) := 1$. Then there exists a polynomial $g(z) := a_4 z^4 + a_3 z^3 + a_2 z^2 + a_1 z + a_0$ where $a_0, a_1, a_2, a_3, a_4 \in \mathbb{R}$ such that for some $w_1, w_2, w_3 \in \mathbb{R}^3$, we have

$$g(\langle w_1, x \rangle) = f_1(x),$$

$$\begin{aligned}g(\langle w_2, x \rangle) &= f_2(x), \\g(\langle w_3, x \rangle) &= f_3(x).\end{aligned}$$

Moreover we have

$$g(z) = -\frac{2}{5}z^4 + \frac{8}{5}Cz^3 + \left(\frac{3}{2} - \frac{12}{5}C^2\right)z^2 + \left(\frac{8}{5}C^3 - 3C\right)z + 1,$$

and $w_1 = [1, 1, C - 1]^\top$, $w_2 = [0, 0, 0]^\top$, $w_3 = [0, 0, 0]^\top$.

Proof. Let $g(z) := -\frac{2}{5}z^4 + \frac{8}{5}Cz^3 + \left(\frac{3}{2} - \frac{12}{5}C^2\right)z^2 + \left(\frac{8}{5}C^3 - 3C\right)z + 1$. Let $w_1 := [1, 1, 1 - C]^\top$, $w_2 := [0, 0, 0]^\top$, $w_3 := [0, 0, 0]^\top$. Then it is clear that we have

$$\begin{aligned}g(\langle w_1, x \rangle) &= f_1(x), \\g(\langle w_2, x \rangle) &= f_2(x), \\g(\langle w_3, x \rangle) &= f_3(x).\end{aligned}$$

Thus we complete the proof. $\square$

Lemma B.3. *Let*

$$f_1(x) := -\frac{2}{5}\langle u, x \rangle^4 + \frac{8}{5}C\langle u, x \rangle^3 + \left(\frac{3}{2} - \frac{12}{5}C^2\right)\langle u, x \rangle^2 + \left(\frac{8}{5}C^3 - 3C\right)\langle u, x \rangle + 1,$$

where $C := \sqrt{\frac{15+\sqrt{65}}{8}}$, $u := [1, 1, C - 1]$. It holds that

$$f_1([x_1, 1, 1]^\top) = -\frac{2}{5}x_1^4 + \frac{3}{2}x_1^2.$$

Proof. Note that $\langle u, [x_1, 1, 1]^\top \rangle = x_1 + C$. Hence

$$\begin{aligned}f_1([x_1, 1, 1]^\top) &= -\frac{2}{5}(x_1 + C)^4 + \frac{8}{5}C(x_1 + C)^3 + \left(\frac{3}{2} - \frac{12}{5}C^2\right)(x_1 + C)^2 + \left(\frac{8}{5}C^3 - 3C\right)(x_1 + C) + 1 \\&= -\frac{2}{5}x_1^4 + \frac{3}{2}x_1^2,\end{aligned}$$

where the first step follows from $\langle u, [x_1, 1, 1]^\top \rangle = x_1 + C$, and the second step follows from expanding the higher-order terms and simplifying it. $\square$

B.2 Extend to Neural Networks

We are now prepared to construct a neural network with polynomial activation functions. By carefully designing the weight matrix, we can ensure that the network converges to different fixed points for different initialization.

Lemma B.4. *Let $W := [w_1, w_2, w_3] \in \mathbb{R}^{3 \times 3}$ be a weight matrix. Let $C = \sqrt{\frac{15+\sqrt{65}}{8}}$ be a constant. Let a polynomial activation function $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined as*

$$g(z) = -\frac{2}{5}z^4 + \frac{8}{5}Cz^3 + \left(\frac{3}{2} - \frac{12}{5}C^2\right)z^2 + \left(\frac{8}{5}C^3 - 3C\right)z + 1,$$

We define one layer of a neural network, denoted as f , as follows:

$$f(x; W) := [g(\langle w_1, x \rangle), g(\langle w_2, x \rangle), g(\langle w_3, x \rangle)]^\top.$$

Then there exists a weight matrix W following statements hold: